

UX Masterclass

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Topics to be covered today

- Understanding User Experience
- Learning Basics of Figma
- Introduction to Shiprocket Wireframe Library

User Experience

**UX = Emotions + Perceptions +
Preferences/Behaviors**

experienced & demonstrated by
users when interacting with a
product, system, or service.

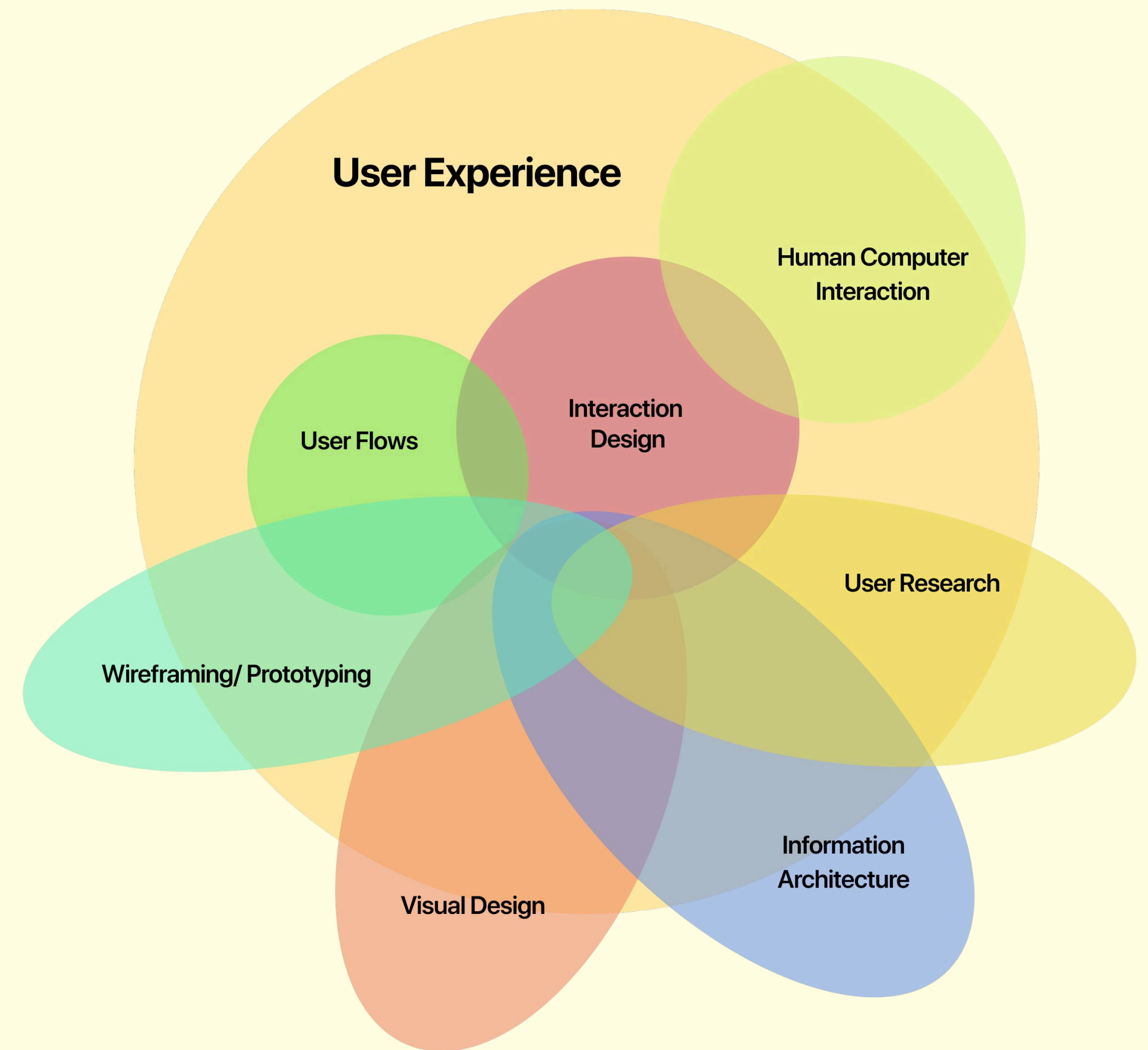


What is UX Design?

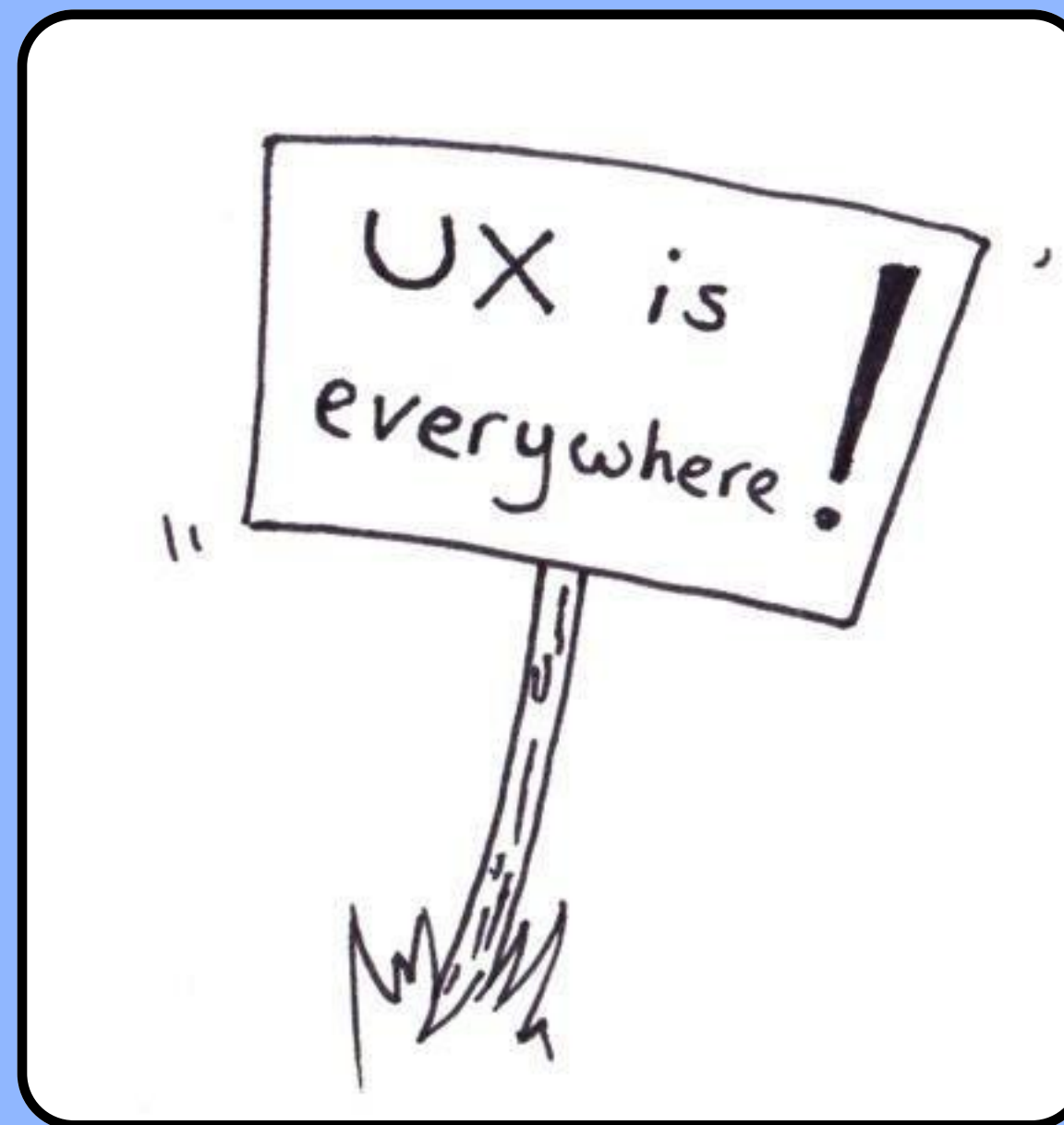
UX design is the process of creating products that aim to provide **meaningful** and **relevant experiences** to users.

The goal of UX design is to create user-centered solutions that are:

- **Intuitive**
- **Efficient**
- **Enjoyable to use.**



User Experience exists all around us





Do I Pull? Do I Push?



How many a fingernail have
been destroyed by this thing?



Where is the edge!?!

Really Apple?



Someone will get hurt...



Where do I go?



Design Fixes that Shine



DESIGNING THE **PRODUCT** VS DESIGNING THE **EXPERIENCE**





Key Takeaways

Prioritize understanding how users feel while using your product, as emotions directly impact:



Overall Success Of The Product Experience



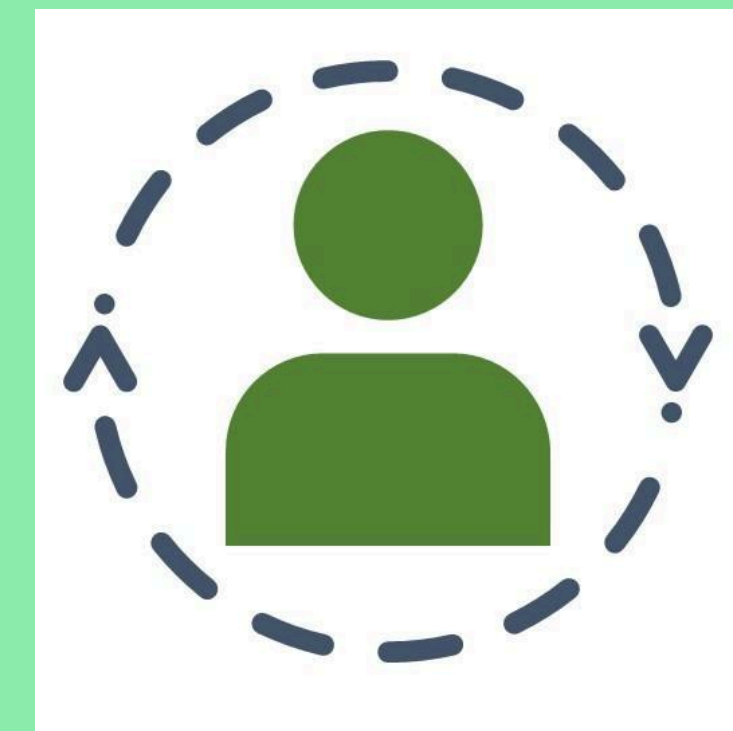
Engagement



Retention



Loyalty



Tenets of Good UX Design

A good UX design is:

- Usable
- Findable
- Accessible
- Useful
- Valuable
- Credible
- Desirable



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Focus for today

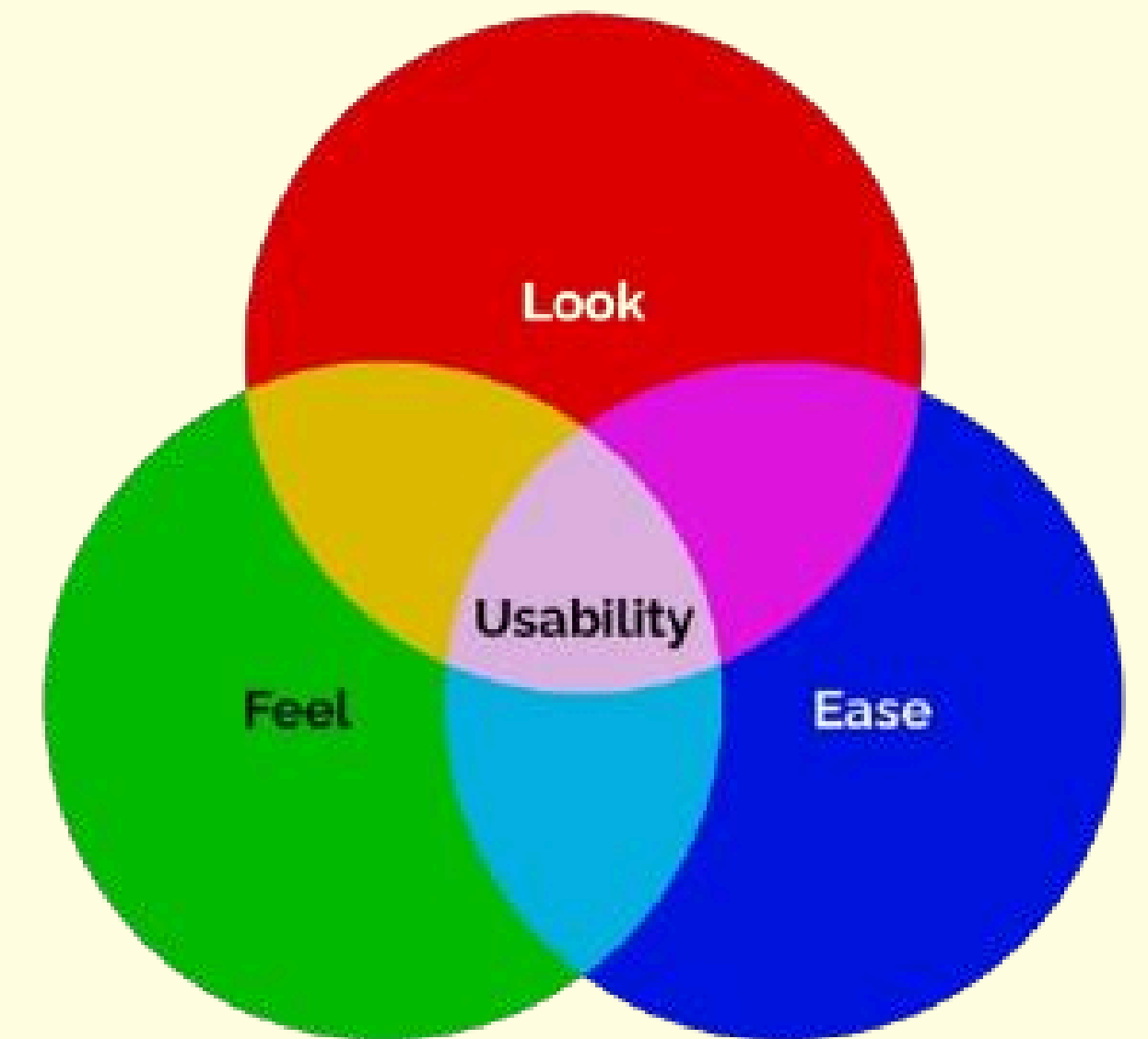


What is Usability?

Usability = **Simplicity + Accessibility of Use**

Refers to what extent:

- » a website or application is **comprehensible** in its presentation
- » it can be used **intuitively** by a new user
- » has an **attractive user interface** design





Control Room of the Nuclear Power Plant

in Three Mile Island, Pennsylvania, USA

- CR is the **epicenter of the operation**.
- Every action and decision echoes through the entire facility.
- Operators monitor complex systems, make crucial decisions, and maintain the delicate balance of nuclear reactions.



What happened?

- In March, 1979, a **failure in the secondary cooling system** (used to cool the steam generators) led to an increase in temperature and pressure inside the reactor.
- To prevent overheating, the **reactor automatically shut down** within seconds, stopping nuclear fission with control rods.
- After the shut down, residual heat continued to build up, which is normal, & needed to be controlled by proper cooling.
- A relief valve, that helps release pressure from the reactor, opened to release pressure, as expected.
- After the initial malfunction and the opening of the relief valve, **multiple alarms got triggered** in the control room
- **The problem:** The valve should have closed after releasing the excess pressure, sealing the system, but it got stuck open.
- This allowed cooling water, which was supposed to keep the reactor cool, to escape, causing the temperature to rise.
- The control room system failed to indicate the open valve, leaving operators unaware of the coolant leak.
- The reactor core reached a temperature of **4300°F**, heating 150 tons of uranium.
- **Ultimately, this led to a partial meltdown of the TMI-2 reactor core, destroying it, and resulting in some radioactivity release.**

Why this really happened?



Overwhelming Alarms: The control room triggered hundreds of alarms simultaneously, making it impossible for operators to identify the critical issues amidst the noise.



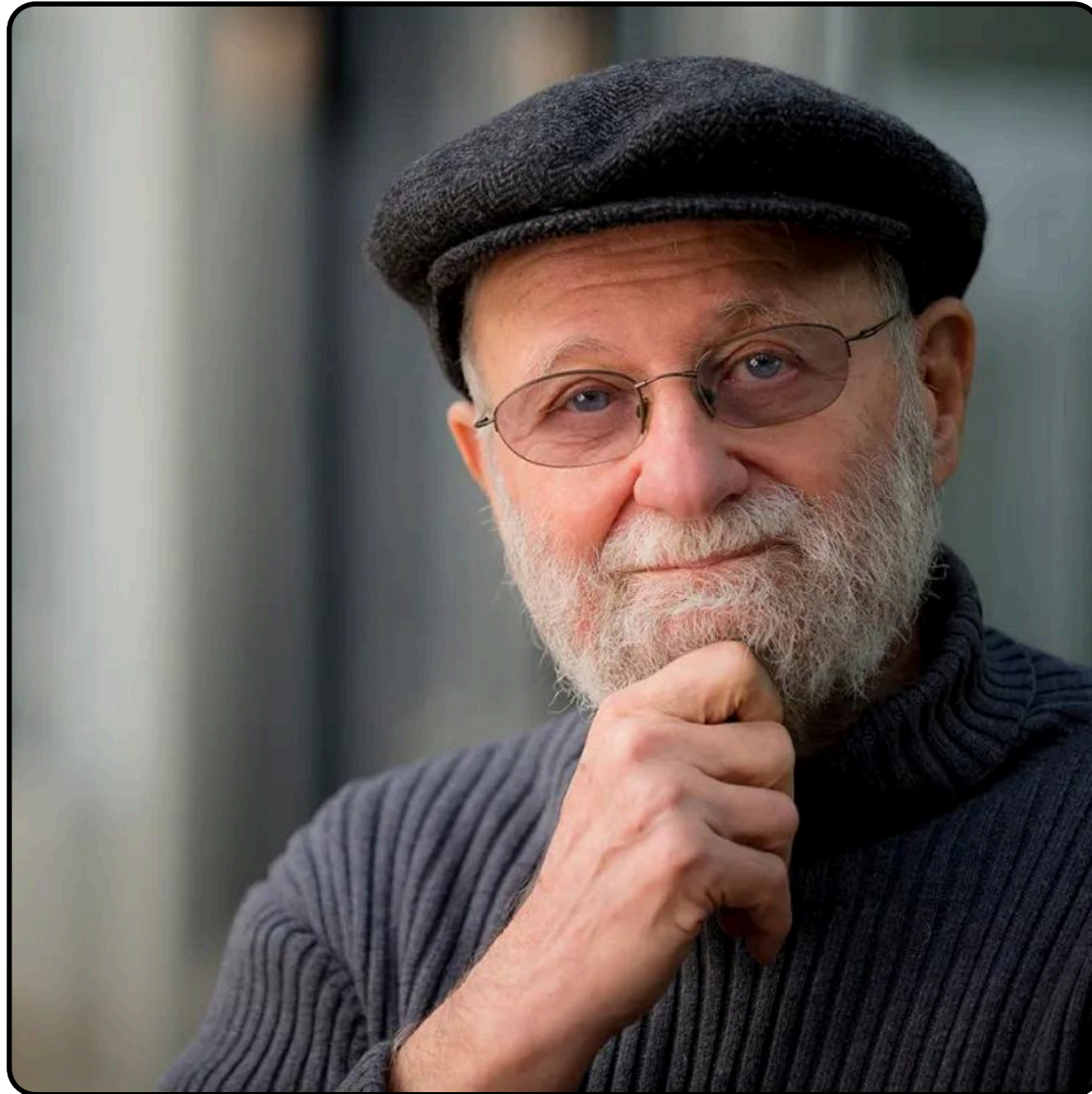
Lack of Prioritization in Warnings: Critical alarms were not differentiated from minor ones, making it hard for operators to quickly assess the severity of the malfunction and respond appropriately.



Cognitive Overload: The sheer number of displays, buttons, and alerts overwhelmed the operators, contributing to incorrect decisions and delays in responding to the core issue.



Misleading Indicators: Critical indicators, like the position of the relief valve, were not clearly visible to operators. The control panels falsely indicated it was closed. **The light for the valve was actually linked to its switch (being turned on/off), & not actually to the valve being open or shut!**



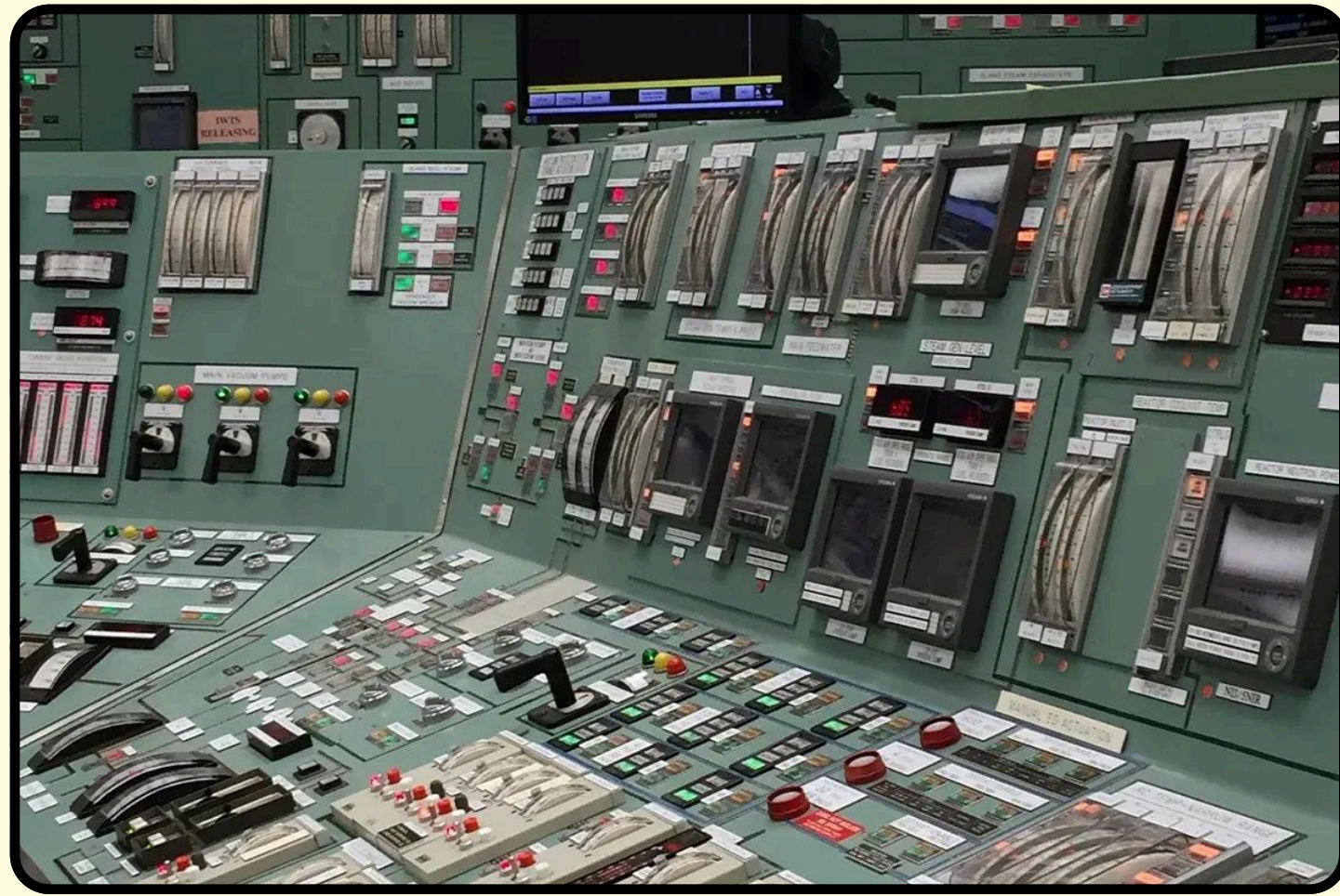
Don Norman, one of the godfathers of UX Design, was one of the experts who investigated this event.

Norman is on record saying,

“The control room and computer interfaces at Three Mile Island could not have been more confusing if they had tried.”

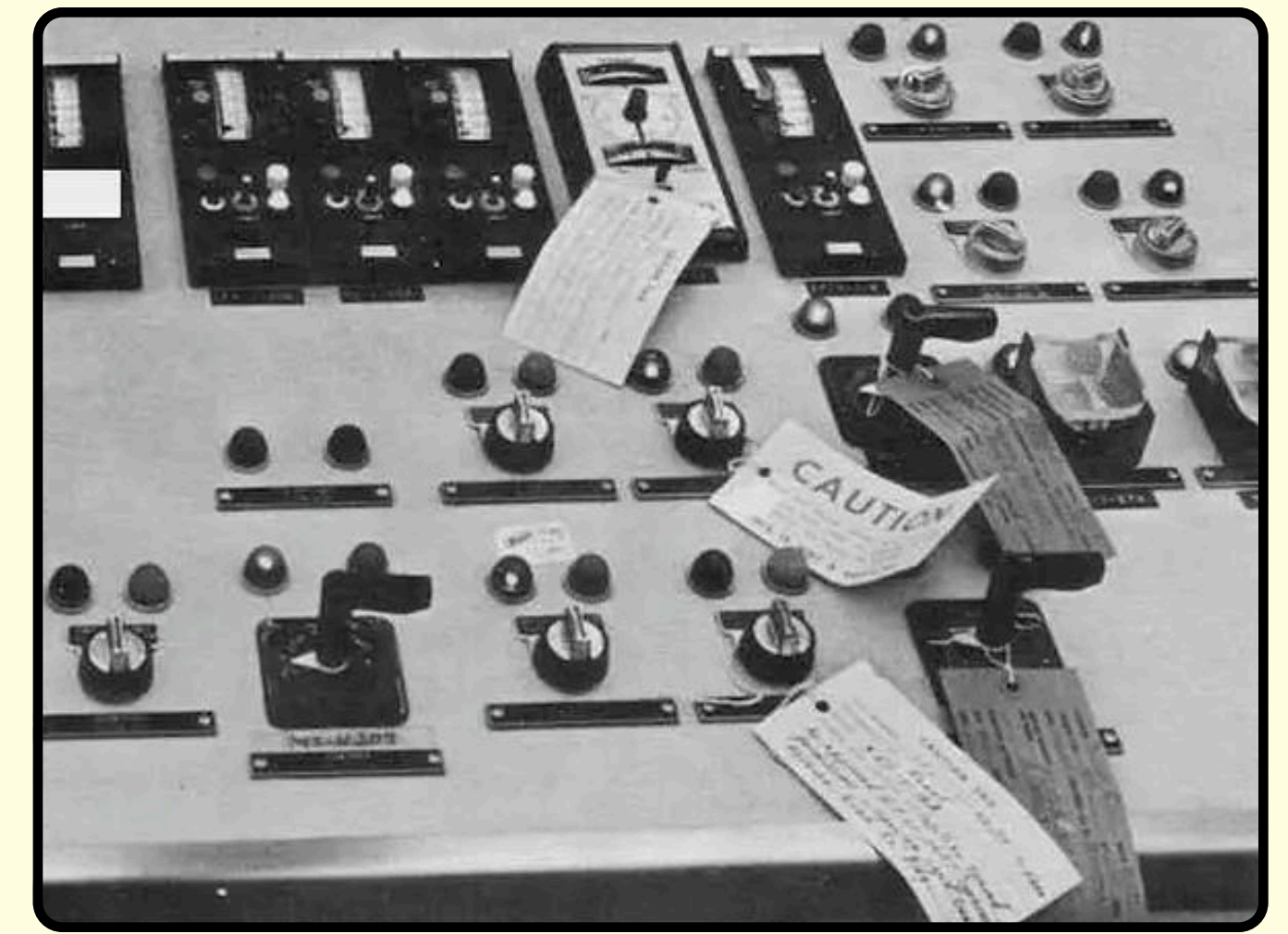
Norman noted that the control panel had the following issues:

- 14 different meanings for red
- 11 different meanings for green
- Lights were not grouped logically
- The plant valve alert and elevator alerts were next to each other
- The release valve did not indicate that the valve was actually closed or open

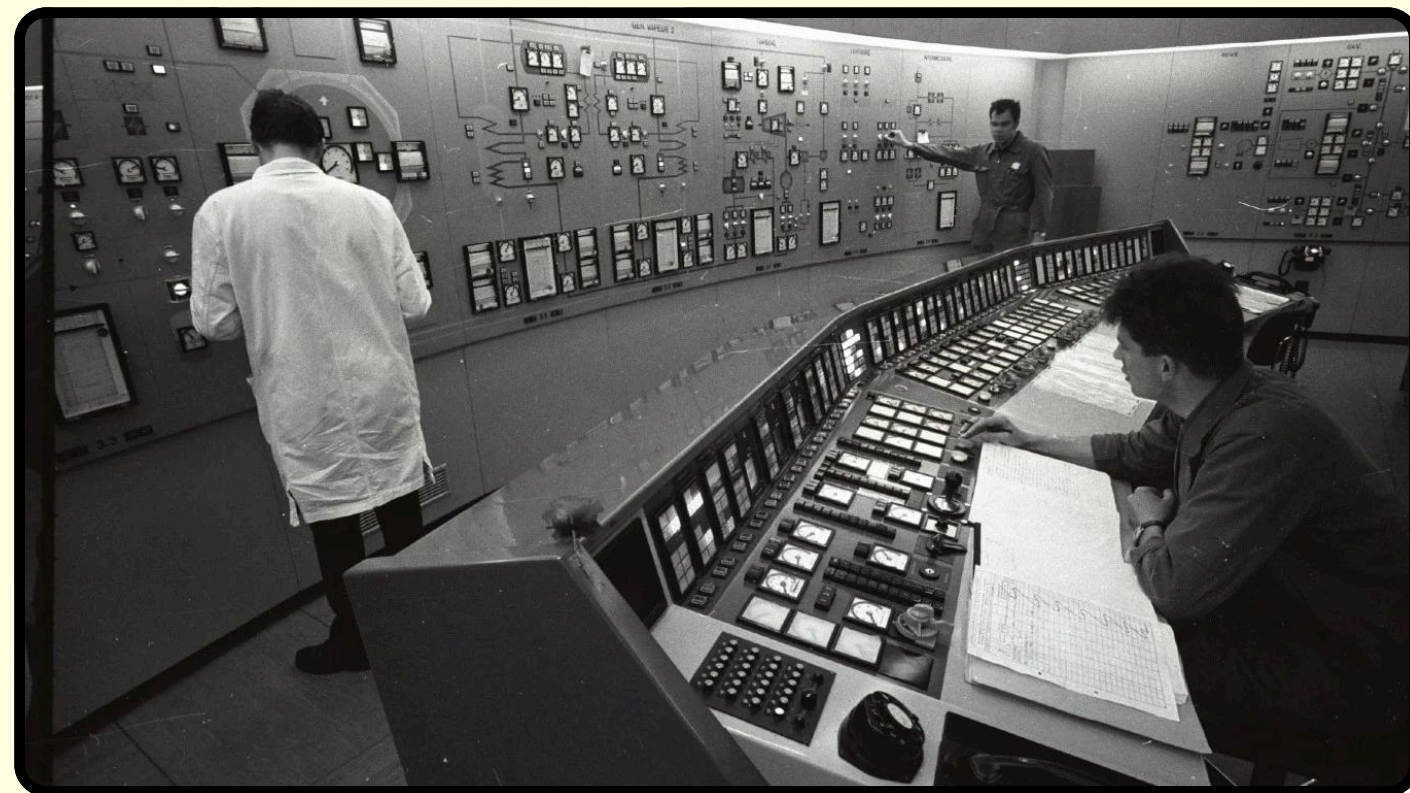


Normal operations were already very complex

Nuclear Emergencies were rare



Even emergencies required a 300 page manual!

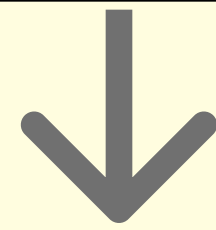


Doing the right thing quickly is essential

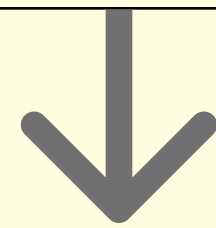


Impact

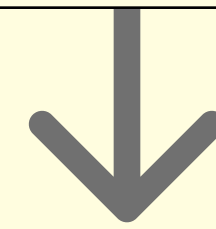
Control room design flaws **hampered the operators' ability** to quickly diagnose and mitigate the escalating reactor issue.



The clean-up of TMI-2 **cost \$973m** and took **12 years** to complete.



Widespread public fear and health concerns followed, which caused a halt in new nuclear power plant construction in the U.S. for decades, despite it being a cleaner alternative for energy fuel.



The event led to **major reforms** in nuclear plant **safety protocols, operator training, and control room designs** to avoid similar confusion in the future.



Key Takeaways

» Good design is always about real people & their situations

Most things are designed for how people are **supposed** to act instead of **how they actually** act. Studying the people and their context brings the problem into focus.

» To design something useful, strip the problem down to what's essential

Reduce the problem to “what needs to be on this screen to help someone decide what to do next”.

» Flow is the most important aspect of design

Allow users to **FLOW** through the steps of diagnosing and solving problems in a natural, quick and friction-less way.

» Designers need to account for real human behavior to mitigate human error

Users are prone to errors, distractions, and misunderstandings. Designers must account for real human behavior to minimize errors and create systems that are intuitive, error-tolerant, and easy to recover from.

Jacob Nielsen's Usability Principles

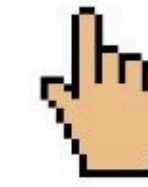
Jacob Nielsen is a usability pioneer who formulated the **10 broad usability principles** that create better user experiences

These heuristics have been created through **observations** and the **expertise** acquired during their years of work experience



Visibility of
System Status

1



Match Between
System & Real World

2



User Control
And Freedom

3



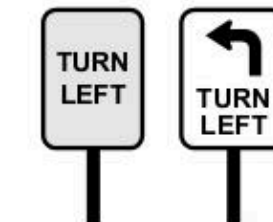
Consistency
And Standards

4



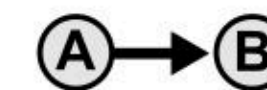
Error
Prevention

5



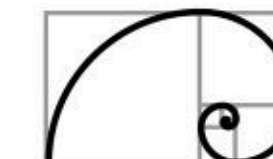
Recognition
Rather Than Recall

6



Flexibility And
Efficiency of Use

7



Aesthetic And
Minimalistic Design

8



Help Users
With Errors

9

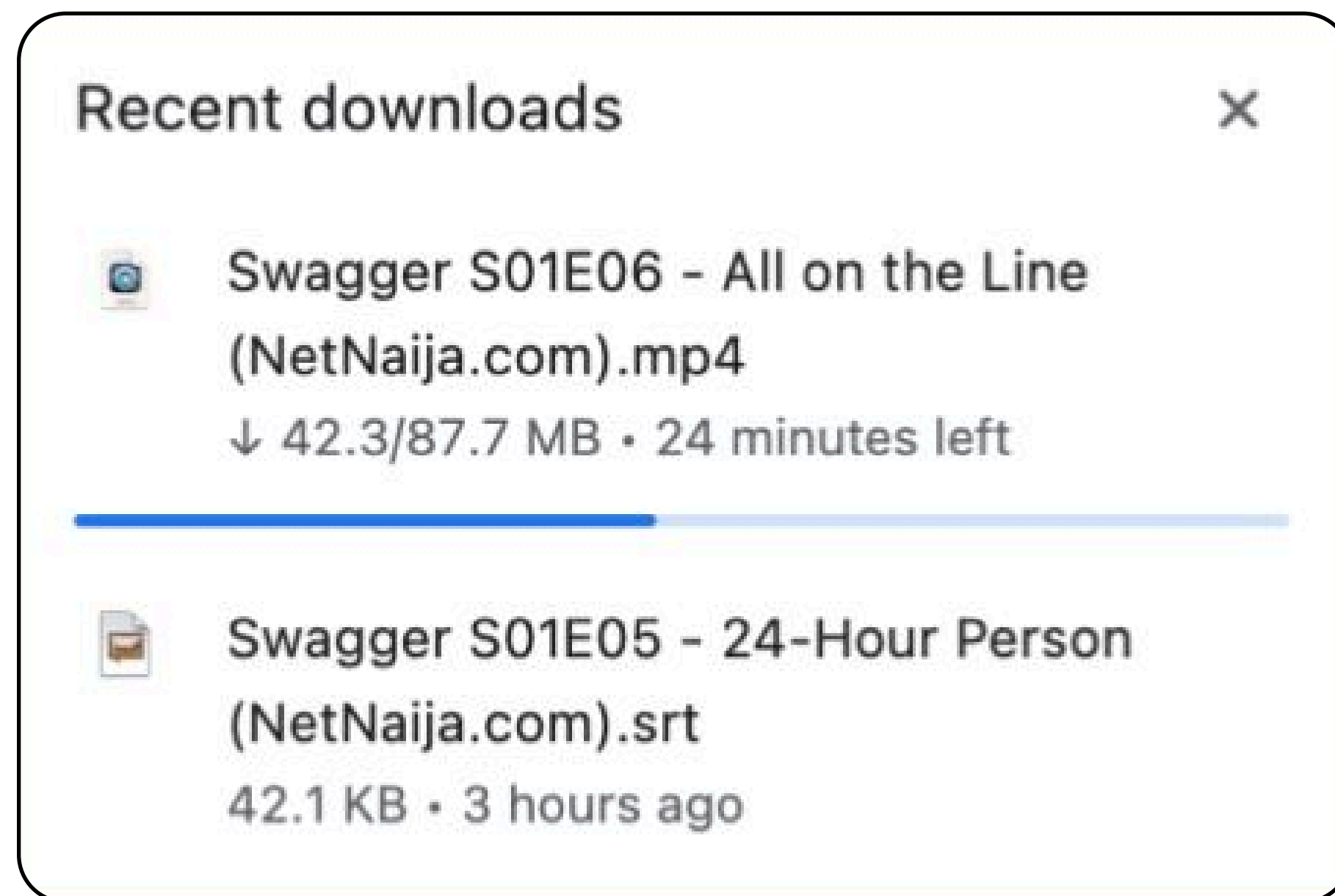


Help And
Documentation

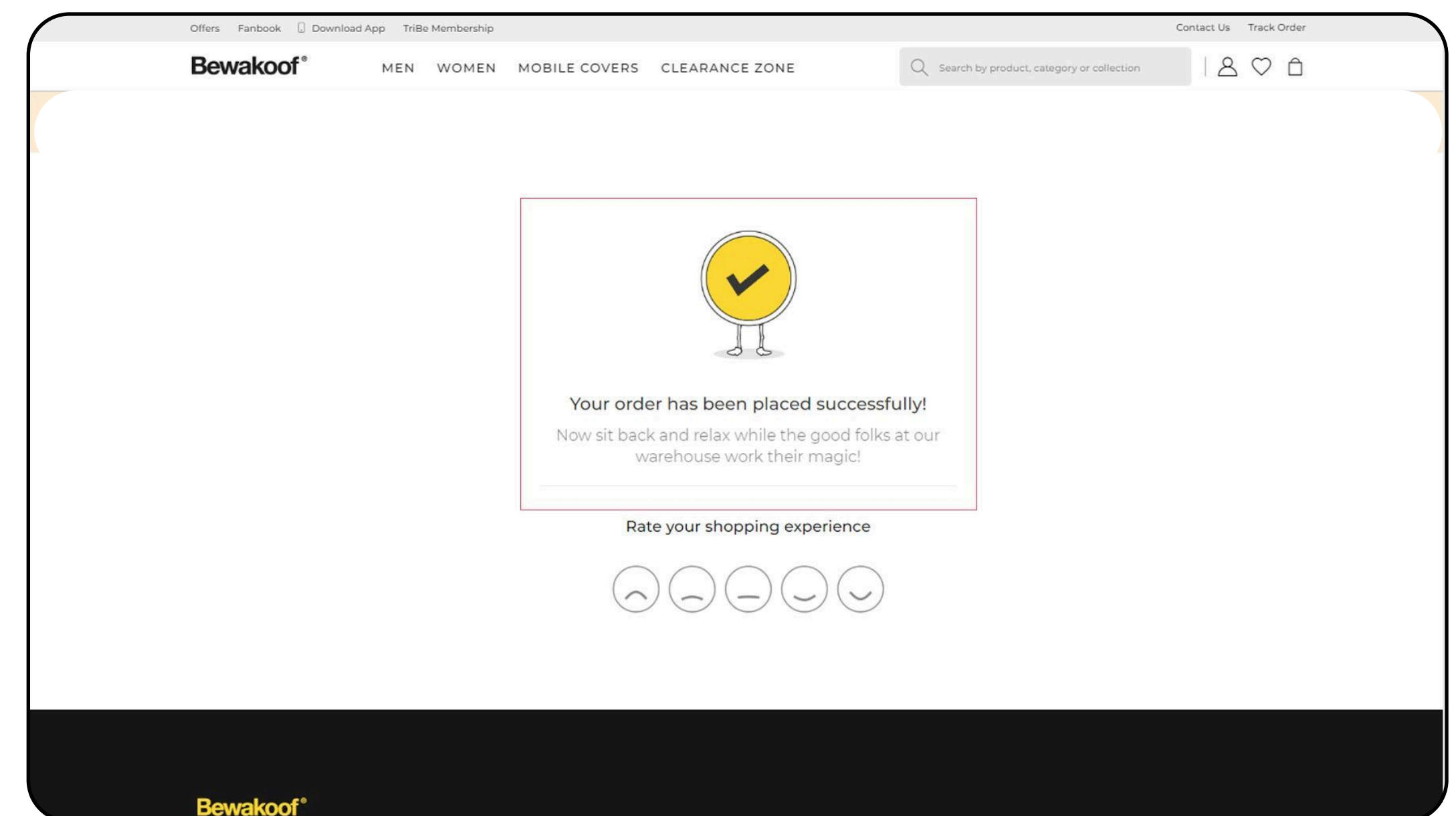
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1. Visibility of System Status

The user should always be informed about what is going on with the system through appropriate feedback.



Visibility of Download Status



Visibility of Order placed

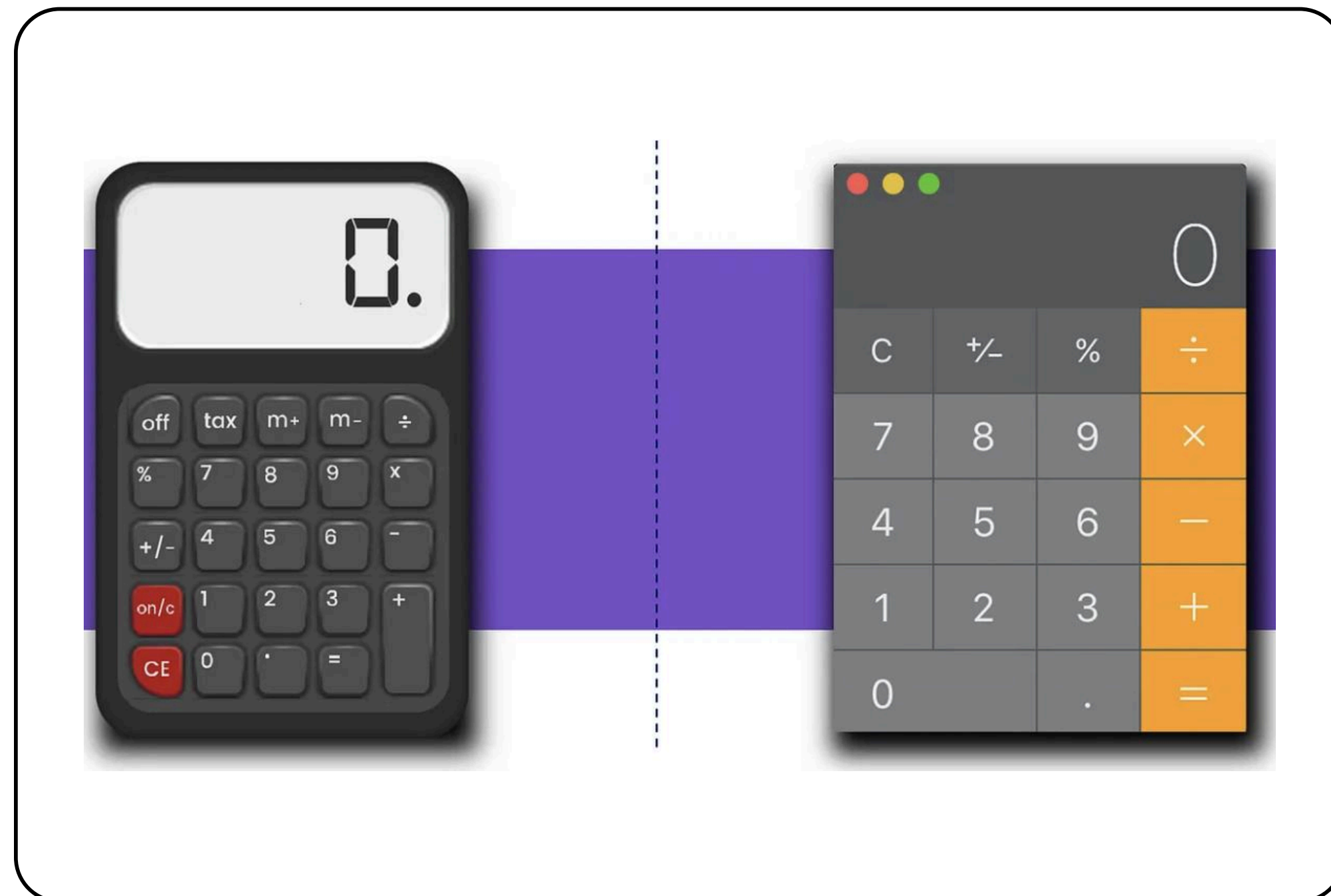
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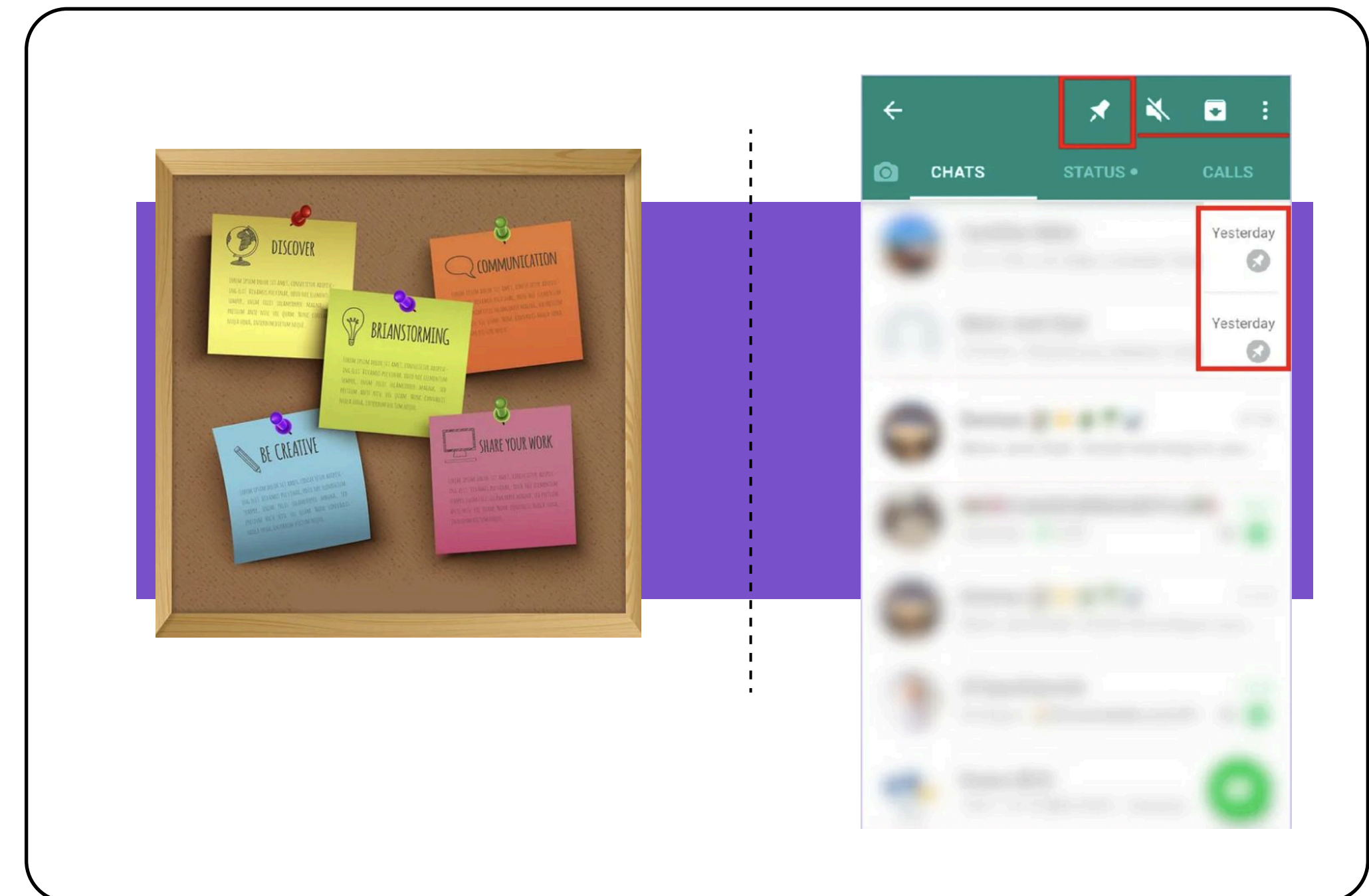


2. Match between system and the real world

The system should speak the user's language,
with words, phrases and concepts familiar to the user.



A physical calculator and mobile calculator



Whatsapp pin feature

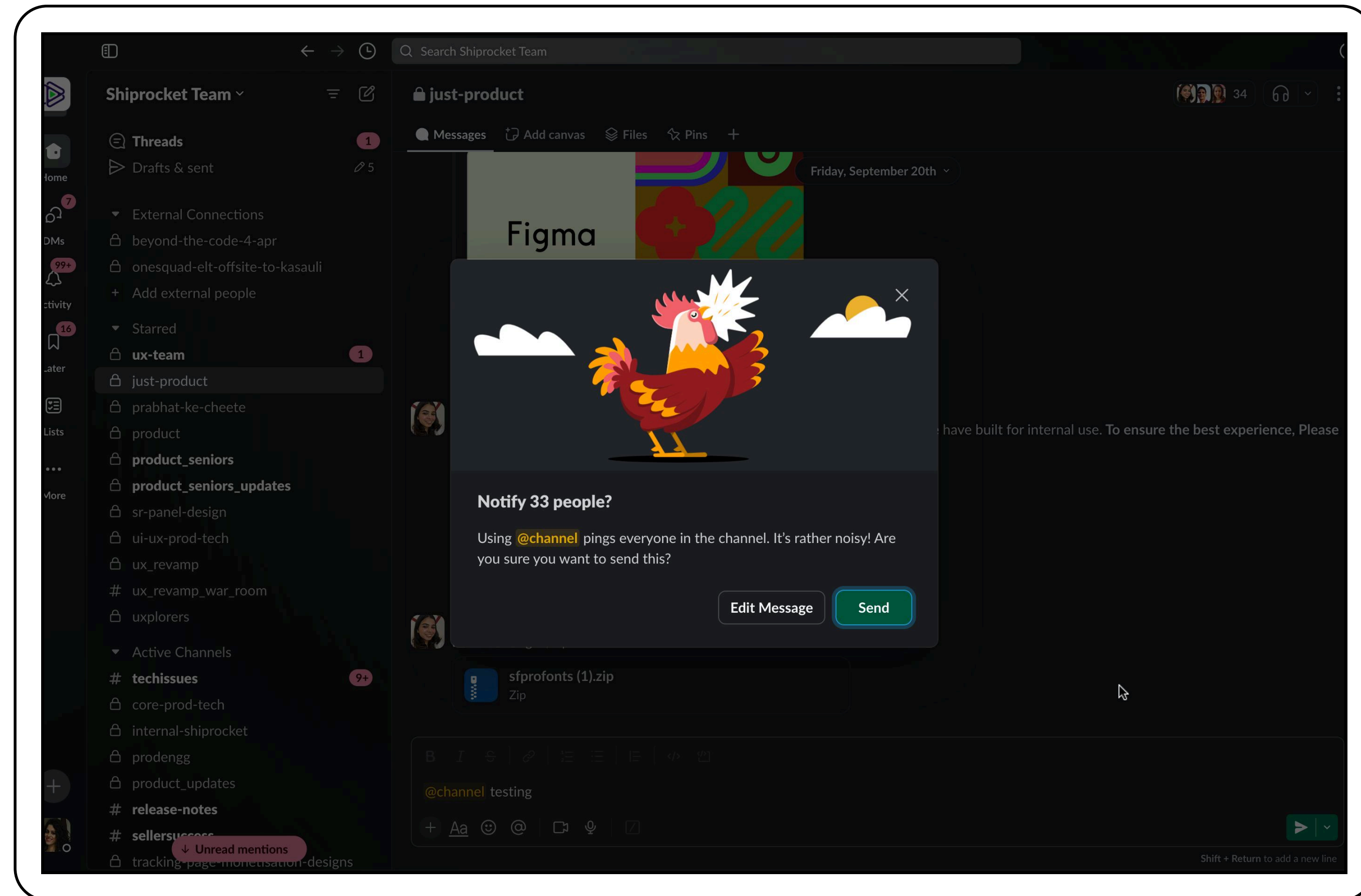
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3. Error preventions

The system should be designed in a way that minimizes the likelihood of errors.



Slack confirms before sending a message to a channel with large number of participants

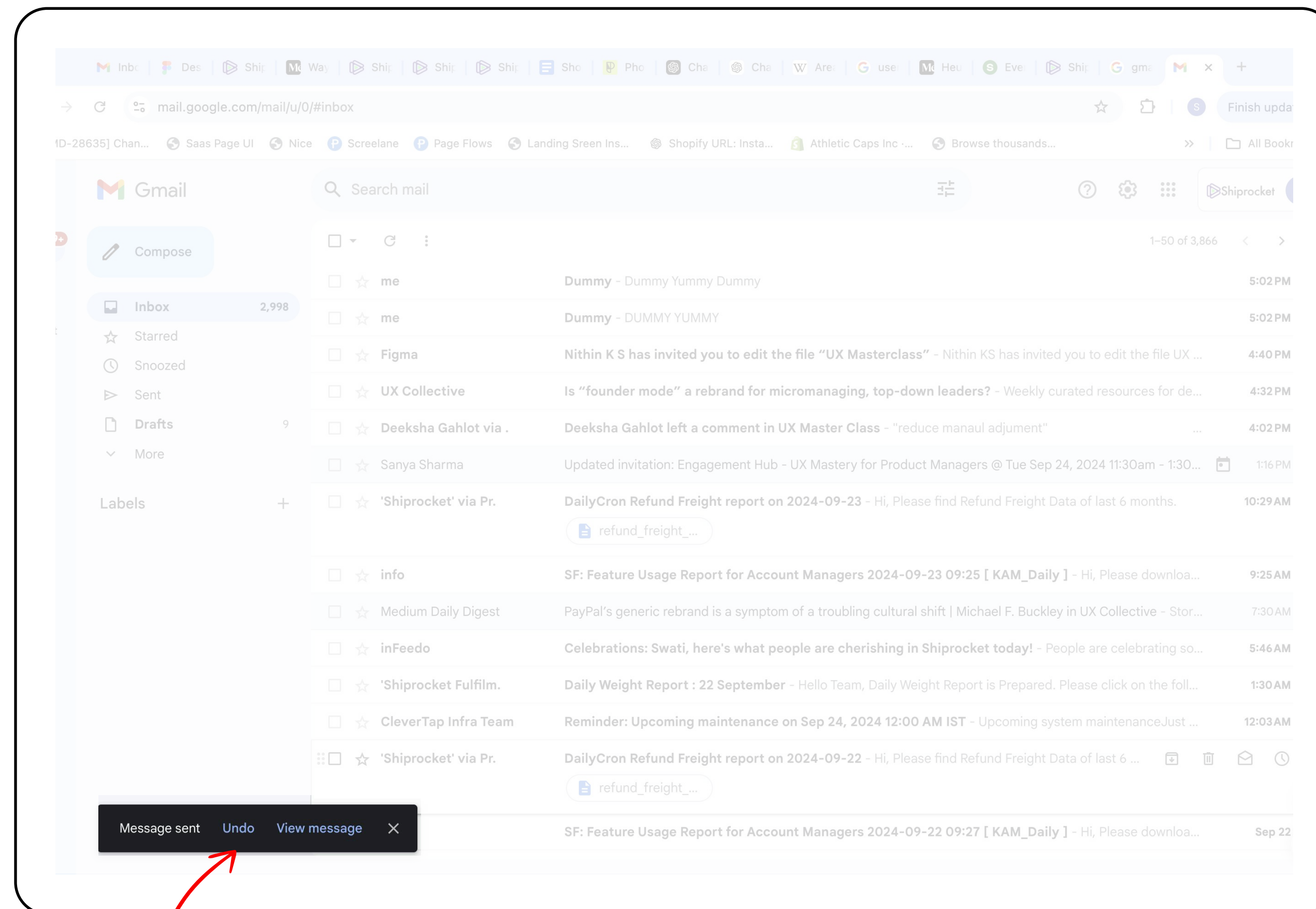
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4. User control and freedom

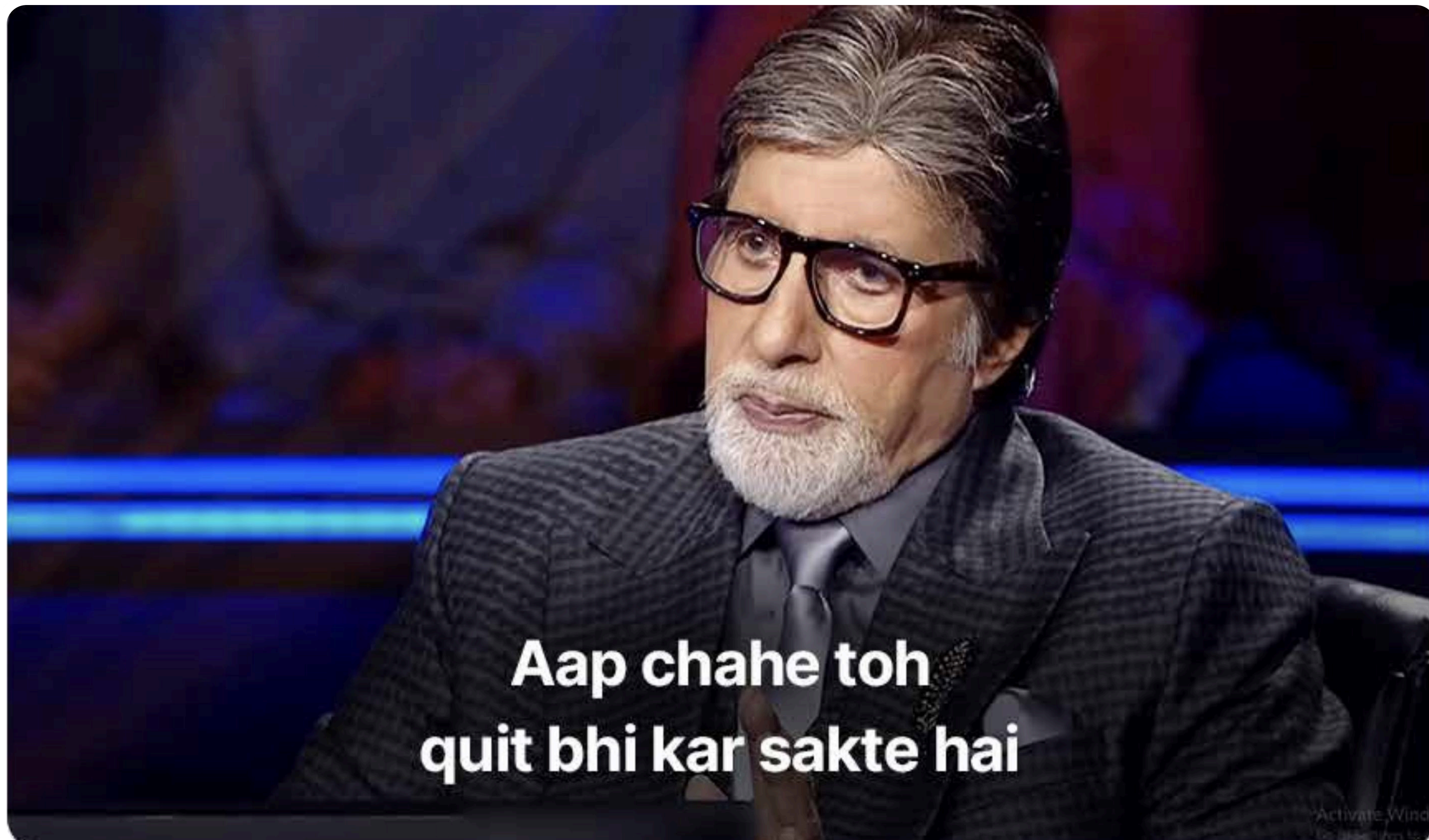
The user should be able to easily navigate and escape from any unwanted situations.



Gmail provides an 'Undo' option to unsend an email after it's sent

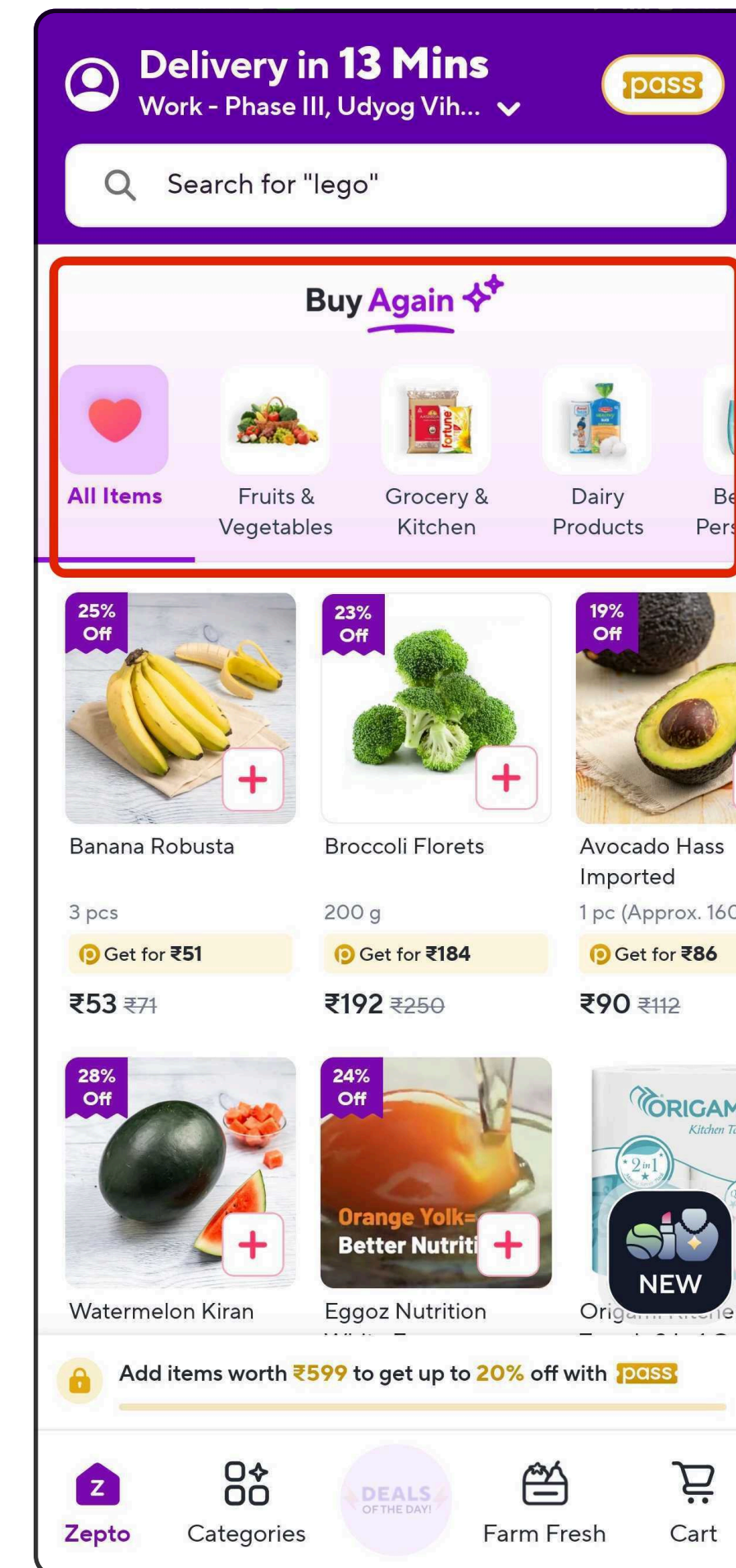
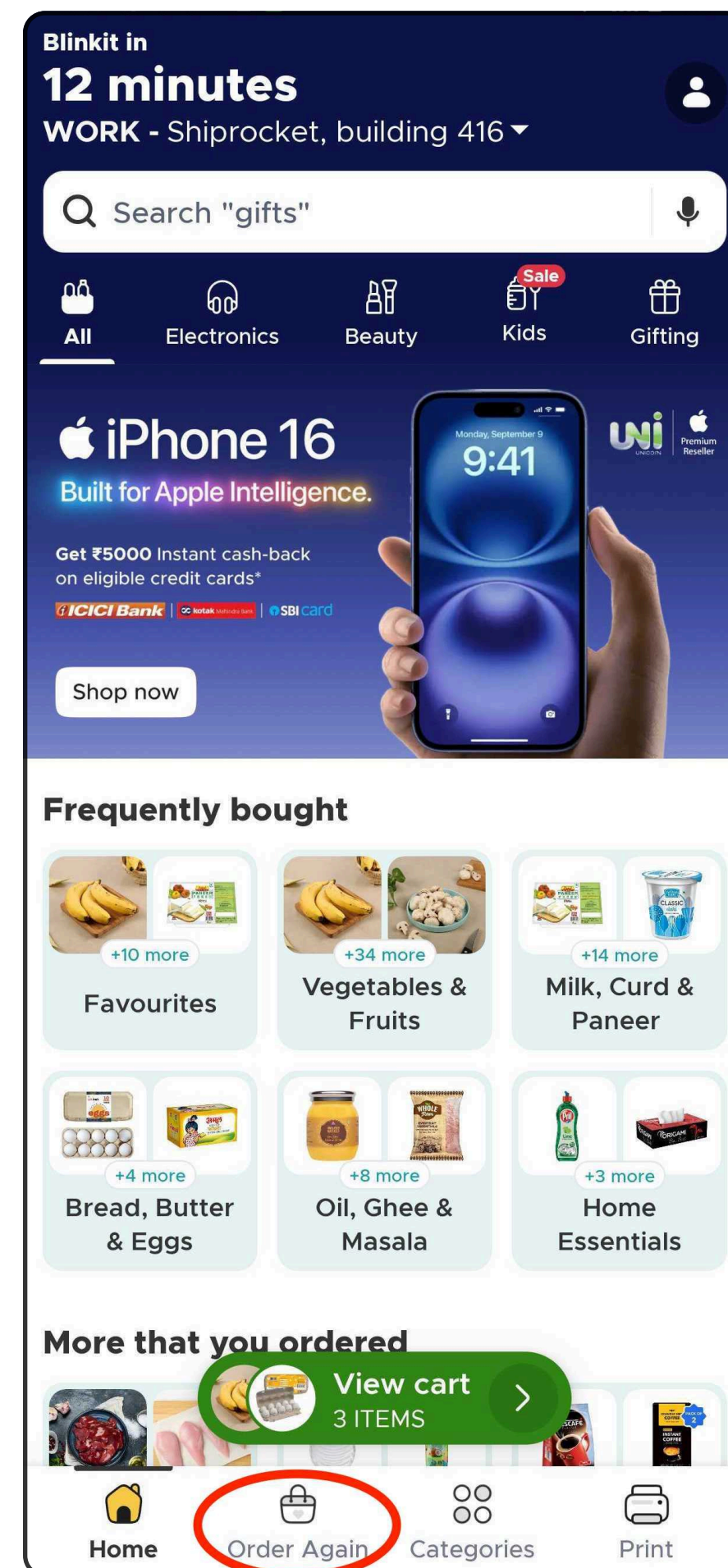
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5. Recognition rather than recall

The system should minimize the user's memory load by making objects, actions, and options visible.



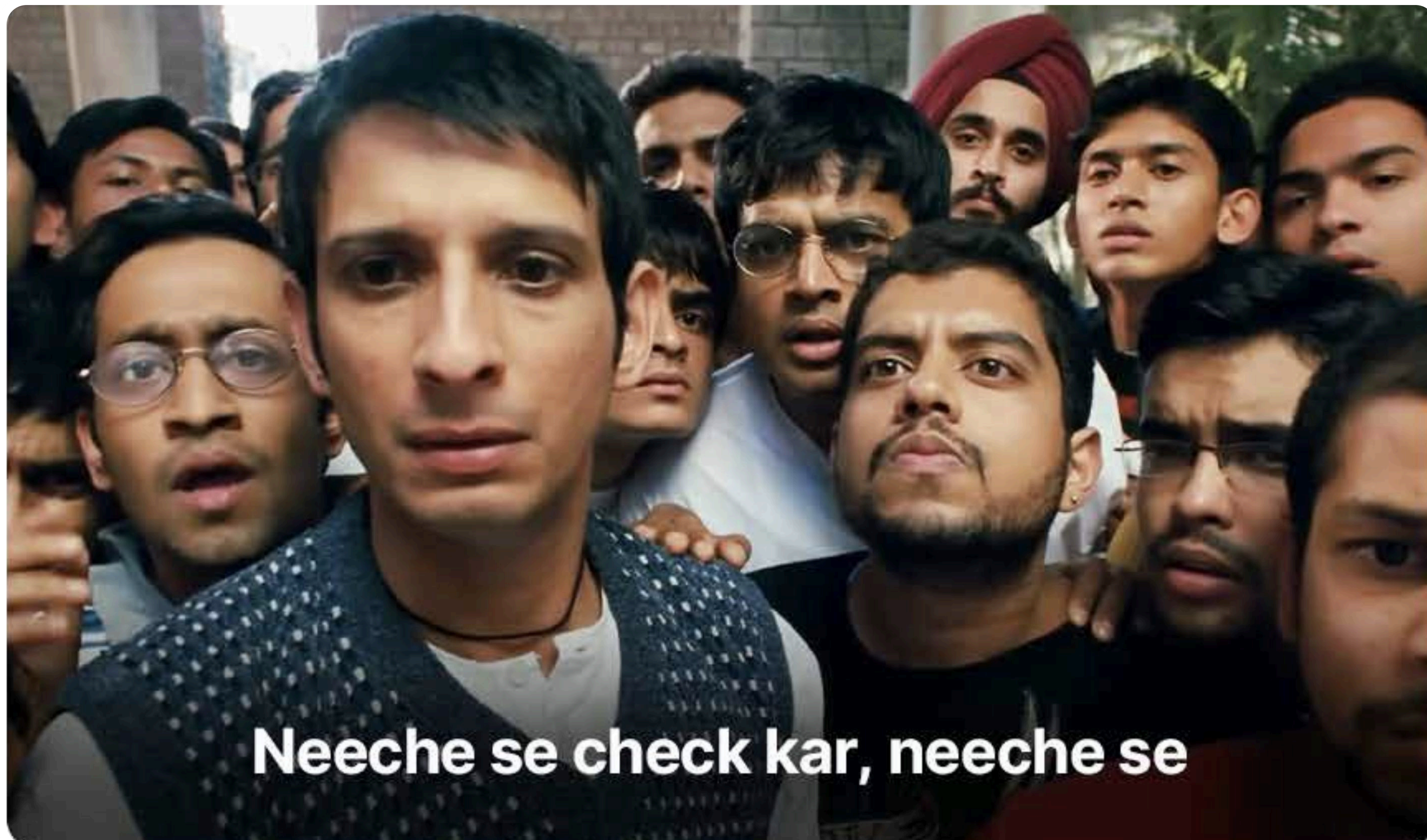
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6. Flexibility and efficiency of use

The system should cater to both novice and expert users with different ways to achieve the same goal.



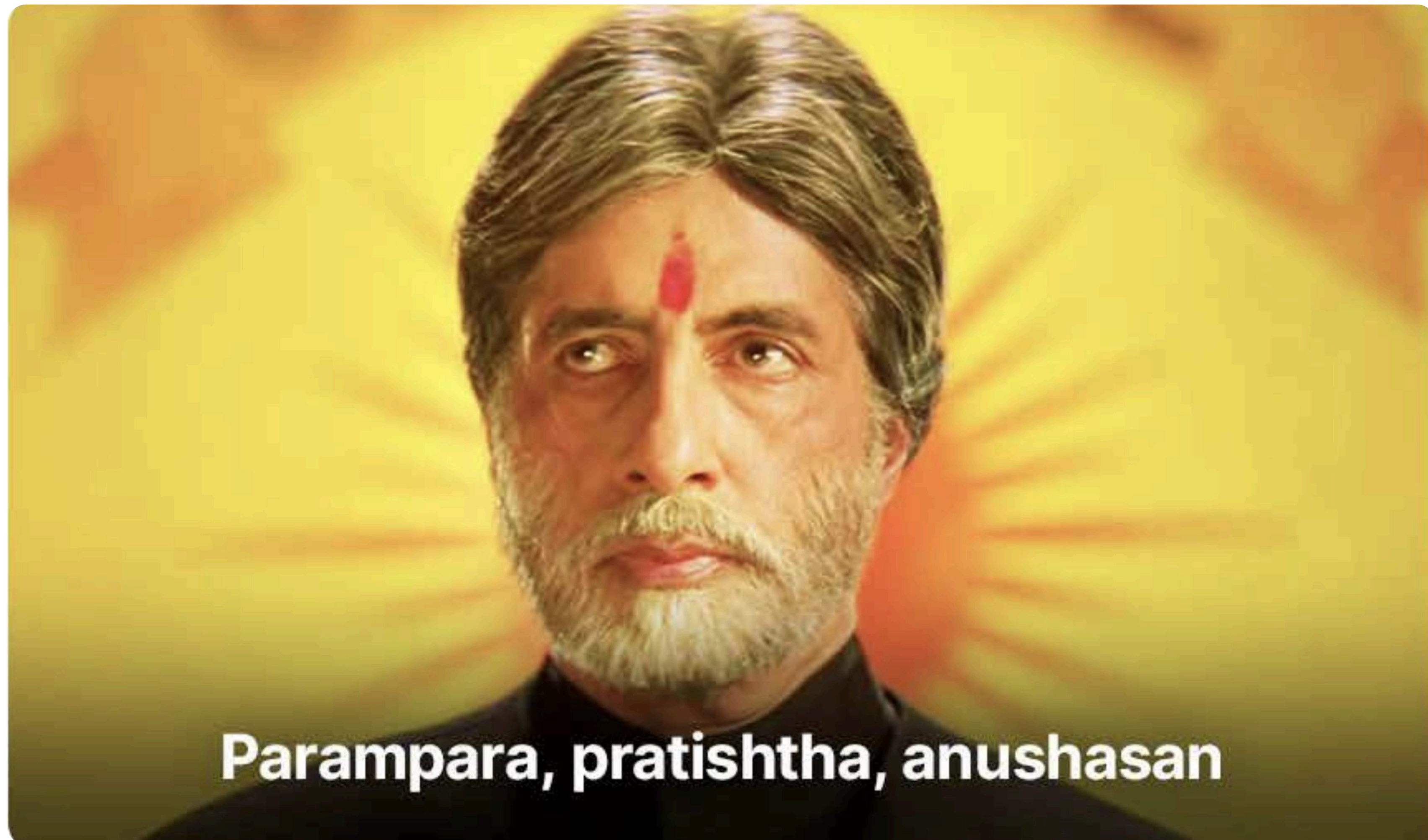
7. Help users recognize, diagnose and recover from errors

The system should provide helpful and clear error messages that suggest solutions to the user.



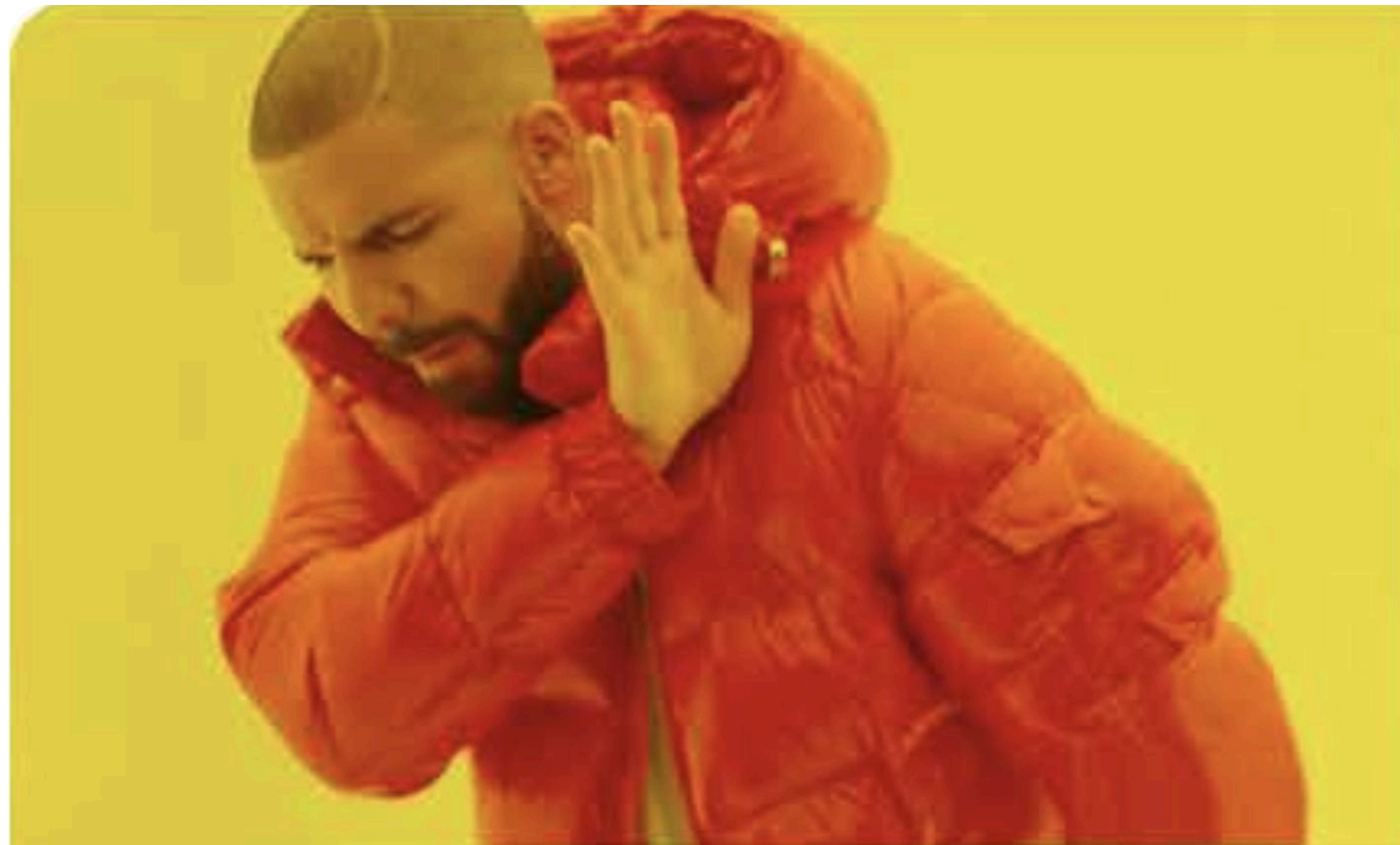
8. Consistency and Standards

The system should follow platform specific conventions and be consistent throughout.



9. Aesthetic and minimalist design

The interface should be designed in a way that is visually appealing, without any extraneous information.



**Crowded design
with no breathing
space**

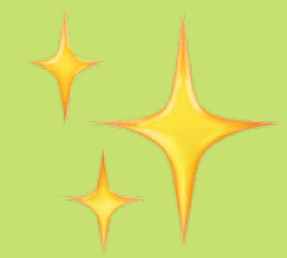


**Aesthetically
pleasing and
minimalist design**

10. Help and Documentation

The system should provide easily accessible documentation and help features.





**Thank you for staying
UX-cited till the end !**

